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Comparison between Weighted Monte Carlo and Generalized Weighted Monte Carlo methods

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Summary

In this work we performed a study comparing the pricing methodologies of the Weighted Monte Carlo (WMC) and Generalized Weighted Monte Carlo (GWMC) methods on call and put options of 3 different underlying assets. First, we introduce the methodologies explaining their differences. Next, we show how to calibrate and apply them. Finally, we compare their performance by means of two tests, from which we conclude that the GWMC has a better chance of reproducing the option prices due to the improved distribution of prior weights that holds more information than the uniform one used in WMC.

Chapter 1

Introduction

In this section we start by introducing the basic concepts that will be handled throughout this document. First, a **financial derivative** is a contract whose value depends on another financial product or variable, for example, an asset, an interest rate, a foreign exchange rate or a commodity. This second financial product is referred to as **underlying asset**. The main derivatives traded in the market are forward contracts, futures and options. In this project we mainly handle options. An **option** is a financial derivative that gives the holder the right, but not the obligation, to purchase or sell from the writer a predetermined asset for an agreed price[7]. Depending on the type of option (European or American) the right can be exercised only at the maturity of the contract or at any time until the maturity. This right can also be bought and sold in the derivatives market or outside it. The purpose of this type of contracts can be seen from two perspectives [11]:

- **Microeconomics**: investors can protect themselves from risk and also speculate.

- **Macroeconomics**: they help make a more efficient market for the values of the underlying assets, they improve the level of liquidity of the market, they expand opportunities of arbitrage, they allow controllable risk profiles and returns.

Option markets were not regulated until 1973 when the Chicago Board Options Exchange was created. Unregulated derivatives markets caused some struggles throughout history, one of the most famous being the "tulipmania"

in the 17th century in Holland; although, long before that, contracts similar to options already existed. In Mesopotamian times, farmers would make contracts in order to sale their future harvests for a fixed price, protecting themselves from price fluctuations. In the Bible[6], more specifically in Genesis 29, Laban offers Jacob an option to marry his younger daughter in exchange of 7 years of labour. As it was not regulated, when Jacob completed his part of the contract, Laban only let him marry his elder daughter Leah. However, at the end, Jacob married both. Also, in Ancient Greek times, Aristotle in his work *Politics* makes a reference to an anecdote of Thales of Miletus in which he successfully speculates with an option contract.

"Tulipmania" is a historical event that occurred in Holland revolving around tulip bulbs[8]. They were so popular that their price skyrocketed and even multiplied by 20 in a three months period in the beginning of 1673. A rare tulip, Semper Augustus, even reached the price of a luxurious house in Amsterdam. The evolution of these prices made the tulip growers buy put options (right to sell) to protect their profit if the price went down, and tulip wholesalers bought call options (right to buy) for the opposite reason. When the tulip bubble burst, the Dutch people had already invested everything they had into the tulips market and lost it all. Since the option market was not regulated, it was impossible to make investors fulfill their obligations specified in the contracts. The Holland economy entered a recession.

Since then, options were not seen with good eyes until the second half of the 20th century. In 1968, Myron Scholes and Fischer Black started looking for a relation between the prices of an option and its underlying asset, and deduced a formula in discrete time. Later, in 1973, Bob Merton published the Black-Scholes formula in continuous time[10]. This model worked wonders: it was able to predict the behaviour of the market under normal circumstances. In 1997, Merton and Scholes received the Nobel Price in economics. Sadly, Myron Black had already passed away. Merton and Scholes started a hedge fund and had wonderful returns the first three years. However, due to a crisis in the Asian markets and another one in Russia, they ended up being rescued by the American Federal Reserve.

After the publishing of the Black-Scholes formula, the popularity of options increased exponentially and, since then, people have tried to improve it and also to come up with new models. In this work we will compare two of them: the Weighted Monte Carlo and the Generalized Weighted Monte Carlo. As their names clearly indicate, the later is a generalization of the former, and was proposed just 3 years ago. The next chapter introduces both

models and explains how to apply them. Next, we present the results obtained and an analysis. In the last chapter we summarize the most important results and indicate possible lines of further investigation.

Chapter 2

Overview of the methods

2.1 Formulation

In this section, we will introduce the formulae for option pricing of the Weighted Monte Carlo method (WMC) described in [2] and the Generalized Weighted Monte Carlo method (GWMC) introduced in [5].

Given a filtered probability space $(\Omega, \mathcal{F}, \mathbf{F}, \mathbb{P})$, a finite time horizon T, and an adapted price process $S = \{S_t\}_{t\in[0,T]}$ to the filtration $\mathcal{F}$, we can understand the set of N paths produced by a Monte Carlo simulation of S_T as a discrete approximation to the distribution of S at time T, starting from the know price S_0 at time $t = 0$. The formulas for calculating the price today of European call (C) and put (P) options are:

$$C(S_0) = e^{-rT}\mathbb{E}^q[\max(S_T - K, 0)], \quad P(S_0) = e^{-rT}\mathbb{E}^q[\max(K - S_T, 0)], \tag{2.1}$$

where T is the maturity of the option, S_T is the price of the underlying asset at maturity, K is the strike, S_0 is the price of the underlying asset at time $t = 0$ and r is the discount rate. Moreover, $\mathbb{E}$ represents the expectation under the risk neutral measure. It is important to note that the evolution of the underlying asset price is governed by a stochastic differential equation. This stochastic model is required to obtain the price of the underlying at maturity S_T from the initial price S_0. Note that S is a stochastic process, so that S_t is a random variable for each time $t \in (0, T]$.

In the Monte Carlo methodology for European options pricing, we first evolve in time by means of a stochastic numerical method (Euler-Maruyama, Milstein, etc) to discretize the stochastic differential equation and obtain a set

of discrete approximated paths representing the evolution of the underlying asset price. In this way, at each time t we have samples of the random variable S_t, particularly at maturity T we get samples of S_T. At these samples, we compute the payoff of the derivative and next the expectation that appears in formulas (2.1) for the call and put European options.

In the usual previously described Monte Carlo technique, all simulated paths have the same probability. However, WMC aims to change the weights (probabilities) of each path. Note that this change in the probabilities is not supposed to be too big. Indeed, the weights will move as little as possible from the uniform distribution, allowing to fit the prices of several options called the benchmark options. Then, one would be able to predict the prices of options with different strikes using the new calibrated weights of the sample paths.

The Generalized Weighted Monte Carlo method gives an additional twist to WMC by asking a simple question: what would happen if the original weights distribution was not uniform? If we suppose that the market tends to large movements, the tail events of our simulated sample paths should have more weight than those near the original price of the underlying asset. This assumption arises from the fact that large price fluctuations occur much too frequently ([9] and [4]). In [5], the authors reweight the tail events using a distortion function that will be discussed later on in this document.

In order to properly reproduce the distribution of possible prices for the underlying asset, a large number of sample paths is required. Let us suppose that we have $N = 10000$ paths and we want to calibrate the weights using WMC with 10 benchmark options. As we have more paths than options, we can change the weights in an infinite number of ways to exactly calibrate the prices of the benchmark options. In [2], they overcome this problem by choosing the measure closest to the uniform in terms of relative entropy, also known as Kullback-Leibler divergence measure [1]. The relative entropy between two discrete probability measures, $\mathbb{P}$ and $\mathbb{Q}$, is given by:

$$D(\mathbb{Q}|\mathbb{P}) = \sum_i q_i \ln\left(\frac{q_i}{p_i}\right). \tag{2.2}$$

Now, let us introduce the WMC and GWMC methods. Assume we have K benchmark options of the underlying S. Moreover, let us assume that no option in the set can be replicated by a portfolio of the other remaining options and that there is no arbitrage opportunity. We simulate N paths from S and compute the stochastic payoffs $(G_1, \ldots, G_K)$, so that we can

build the payoff matrix

$$G = \begin{pmatrix} g_{1,1} & g_{1,2} & \cdots & g_{1,K} \\ g_{2,1} & g_{2,2} & \cdots & g_{2,K} \\ \vdots & \vdots & \ddots & \vdots \\ g_{N,1} & g_{N,2} & \cdots & g_{N,K} \end{pmatrix}, \tag{2.3}$$

where g_{ik} is the payoff from option k when path i is realized.

If the simulated paths are assigned prior weights given by $\mathbf{p} = (p_1, \ldots, p_N)$, the new weights $\mathbf{q} = (q_1, ..., q_N)$ are calculated as the solution of the following minimization problem:

$$\min_q D(\mathbf{q}|\mathbf{p}) = \min_q \left\{ \sum_{i=1}^{N} q_i \ln\left(\frac{q_i}{p_i}\right) \right\}, \tag{2.4}$$
$$s.t.\ \pi_k = \sum_{i=1}^{N} q_i g_{ik},\ k = 1, ..., K,$$

where π_k represents the market price of benchmark option k. If the prior weights are uniform as in WMC, then problem (2.4) simplifies to:

$$\min_q D(\mathbf{q}|\mathbf{p}) = \min_q \left\{ \ln\left(\frac{1}{N}\right) + \sum_{i=1}^{N} q_i \ln(q_i) \right\} = \min_q \left\{ \sum_{i=1}^{N} q_i \ln(q_i) \right\}, \tag{2.5}$$
$$s.t.\ \pi_k = \sum_{i=1}^{N} q_i g_{ik},\ k = 1, ..., K.$$

Next, by using the Lagrange multiplier approach, the problem (2.4) can be rewritten as the following min max problem (dual problem):

$$q_i = \frac{p_i}{Z(\lambda)} \exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k \right) \quad Z(\lambda) = \sum_{i=1}^{N} p_i \exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k \right). \tag{2.6}$$

In WMC with the prior uniform distribution it would simplify to:

$$q_i = \frac{1}{NZ(\lambda)} \exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k \right) \quad Z(\lambda) = \frac{1}{N} \sum_{i=1}^{N} \exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k \right). \tag{2.7}$$

By using expression (2.6), from (2.4) we can get to the following minimization problem:

$$\min_\lambda \left\{ \ln(Z(\lambda)) - \sum_{k=1}^{K} \pi_k \lambda_k \right\} = \min_\lambda W(\lambda). \tag{2.8}$$

2.2 Solving the minimization problem of the benchmark options

The minimization problem (2.8) requires numerical methods to be solved. In [2], they used the L-BFGS optimization numerical method. In 2006, a more robust, fast, and easy-to-implement calibration algorithm was introduced by Elices and Giménez in [3]. It is an iterative method based on the second-order Taylor approximation of $W(\lambda)$ around an initial point $\hat{\lambda}$:

$$W(\lambda) \approx W(\hat{\lambda}) + \nabla_\lambda W(\hat{\lambda})(\lambda - \hat{\lambda}) + \frac{1}{2}(\lambda - \hat{\lambda})^T \mathbf{J}(\hat{\lambda})(\lambda - \hat{\lambda}). \tag{2.9}$$

So, we need to compute the first and second derivatives of $W(\lambda)$:

$$\frac{\partial W(\lambda)}{\partial \lambda_b} = \frac{1}{Z(\lambda)} \sum_{i=1}^{N} g_{ib} p_i \exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k\right) - \pi_b \tag{2.10}$$

$$\begin{aligned}
\mathbf{J}_{ab}(\lambda) = \frac{\partial^2 W(\lambda)}{\partial \lambda_a \partial \lambda_b} &= \frac{\partial}{\partial \lambda_a}\left[\frac{1}{Z(\lambda)} \sum_{i=1}^{N} g_{ib} p_i \exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k\right)\right] - \frac{\partial \pi_b}{\partial \lambda_a} \\
&= -\frac{1}{Z(\lambda)^2}\frac{\partial Z(\lambda)}{\partial \lambda_a} \sum_{i=1}^{N} g_{ib} p_i \exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k\right) \\
&\quad + \frac{1}{Z(\lambda)} \sum_{i=1}^{N} g_{ib} p_i \frac{\partial}{\partial \lambda_a}\left[\exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k\right)\right] \\
&= \frac{1}{Z(\lambda)} \sum_{i=1}^{N} g_{ib} p_i g_{ia} \exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k\right) \\
&\quad - \frac{1}{Z(\lambda)^2} \sum_{i=1}^{N} p_i g_{ia} \exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k\right) \sum_{i=1}^{N} g_{ib} p_i \exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k\right) \\
&= \sum_{i=1}^{N} g_{ib} g_{ia} \left(\frac{1}{Z(\lambda)} p_i \exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k\right)\right) \\
&\quad - \sum_{i=1}^{N} g_{ia} \frac{1}{Z(\lambda)} p_i \exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k\right) \sum_{i=1}^{N} g_{ib} \frac{1}{Z(\lambda)} p_i \exp\left(\sum_{k=1}^{K} g_{ik}\lambda_k\right) \\
&= \sum_{i=1}^{N} g_{ib} g_{ia} q_i - \sum_{i=1}^{N} g_{ia} q_i \sum_{i=1}^{N} g_{ib} q_i
\end{aligned}$$

$$\mathbf{J}_{ab}(\lambda) = \sum_{i=1}^{N} g_{ib} g_{ia} q_i - \sum_{i=1}^{N} g_{ia} q_i \sum_{i=1}^{N} g_{ib} q_i \tag{2.11}$$

Equation (2.11) is the same as Equation 13 in [3]. In [3], the notation of p and q are different as the calculations are presented for WMC, while here we have generalized them for GWMC. The only difference resides in how q_i is computed. The gradient and the vector $\lambda - \hat{\lambda}$ are:

$$\nabla_\lambda W(\hat{\lambda}) = \begin{pmatrix} \mathbb{E}^q(\mathbf{g}_1) - \pi_1 \\ \vdots \\ \mathbb{E}^q(\mathbf{g}_N) - \pi_K \end{pmatrix} \quad \lambda - \hat{\lambda} = \begin{pmatrix} \lambda_1 - \hat{\lambda}_1 \\ \vdots \\ \lambda_K - \hat{\lambda}_N \end{pmatrix}. \tag{2.12}$$

The minimization of the second order approximation has an analytic solution:

$$\nabla_\lambda W(\hat{\lambda}) + \mathbf{J}(\hat{\lambda})(\lambda - \hat{\lambda}) = 0, \tag{2.13}$$

$$\lambda = \hat{\lambda} - \mathbf{J}^{-1}(\hat{\lambda}) \nabla_\lambda W(\hat{\lambda}). \tag{2.14}$$

Equation (2.13) shows the optimality conditions of this problem, which are obtained by setting the gradient vector of Equation (2.9) to zero. Equation (2.14) shows the partial solution of the second-order approximation. In general, this solution will not be the λ that minimizes $W(\lambda)$, but it provides a new point from which we can obtain another partial solution. This leads to an iterative process that will terminate when the gradient in (2.12) equals (or is close enough to) 0; it will stop when the predicted values of the benchmark options match their market values.

As the Jacobian $\mathbf{J}$ is a covariance matrix, it is semi-positive definite and there is always a minimum. Furthermore, notice that $W(\lambda)$ is convex. Now, if we start far from the minimum, the partial solution of (2.14) can differ significantly from the solution of the previous step, so that the approximation may not be valid anymore. In order to overcome this issue, in [3] the authors propose an under-relaxed Newton update by a step shortening factor α:

$$\lambda = \hat{\lambda} - \alpha \mathbf{J}^{-1}(\hat{\lambda}) \nabla_\lambda W(\hat{\lambda}). \tag{2.15}$$

They also propose starting with $\alpha = 0.01$ and doubling its value after each iteration until it reaches $\alpha = 1$. Also, if the condition number of the Jacobian increases by a factor greater than 10, α should be divided by 5 in order to reduce the step.

Now, we are in a position to write the minimization algorithm for the benchmark options:

1. Simulate N paths of the asset with regular Monte Carlo.
2. Calculate the discounted payoffs of all the benchmark options for each simulated path and create the payoff matrix G.
3. Fix the initial value for the Lagrange multipliers $\hat{\lambda}$ equal to 0, which corresponds to a uniform distribution of weights.
4. Calculate the gradient vector and the Jacobian with Equations (2.12) and (2.11), with the weights previously calculated with Equations (2.7) for WMC and (2.6) for GWMC (WMC could also be computed with this equation setting $p_i = \frac{1}{N},\ i = 1, ..., N$, but the computational time is greater so it is not recommended).
5. Calculate the optimum for the second-order approximation using Equation (2.15).
6. Go to step 4 until the gradient is close enough to zero.

2.3 General Weighted Monte Carlo (GWMC)

2.3.1 Distortion function

How do we choose the original (prior) weights of our simulated paths? How do we account for the fact that the tails are thick? As mentioned earlier, in [5], the authors reweight the tail events using a distortion function of the form:

$$v(P) = \exp\left\{-\gamma\left(-\ln(P)\right)^{\delta}\right\}. \tag{2.16}$$

Actually, depending on being on the loss or gain domain they consider different parameters γ and δ. More precisely:

$$v^{+}(P) = \exp\left\{-\gamma^{+}\left(-\ln(P)\right)^{\delta^{+}}\right\},\ v^{-}(P) = \exp\left\{-\gamma^{-}\left(-\ln(P)\right)^{\delta^{-}}\right\}, \tag{2.17}$$

where v^{+} is used for the gain domain and v^{-} for the loss domain. The parameters γ^{+}, δ^{+}, γ^{-} and δ^{-} will be determined by another minimization problem, see section 2.3.2.

Now, we have to calculate the prior weights using (2.17). In the GWMC paper, the functions v^{-} and v^{+} are defined as $v^{-} : [0,1] \to [0,1]$ and $v^{+} : [0,1] \to [0,1]$, respectively, as two strictly increasing functions satisfying

$v^-(0) = v^+(0) = 0$ and $v^-(1) = v^+(1) = 1$. After some calculations, the following formula to compute the prior weights is obtained:

$$p_i = \begin{cases} v^+\left(\frac{n-i}{N}\right) - v^+\left(\frac{n-i-1}{N}\right), \text{ if } 0 \leq i \leq n, \\ v^-\left(\frac{m-i}{N}\right) - v^-\left(\frac{m-i-1}{N}\right), \text{ if } -m \leq i \leq 0, \end{cases} \tag{2.18}$$

where $v^+(\cdot)$ and $v^-(\cdot)$ are the distortion functions in (2.17). N is the number of simulated paths, and these paths are ordered in ascending order as $-m, -m+1, \ldots, n-1, n$, where path $i = 0$ is the closest to realizing no change in value for the underlying asset.

Note that the expressions in (2.18) are incorrect in the original paper [5]. If we take $i = n$, we would have to calculate $v^+\left(\frac{-1}{N}\right)$, which has a negative argument outside the function domain (so that we would have to calculate the logarithm of a negative number, and we need the weights to be real numbers). Similarly, taking $i = -m$ gives us $v^-\left(\frac{2m}{N}\right)$, which could result in an argument greater than 1, thus falling outside the interval $[0, 1]$, if the number of paths below the original asset price is greater than $N/2$.

During the development of this work, we contacted Hilmar Gudmundsson and David Vyncke, the authors of the paper [5], and raised these issues, to which they kindly confirmed and sent us the corrections. Following their indications, the correct formula to calculate the prior weights is:

$$p_i = \begin{cases} v^+\left(\frac{n-i}{N}\right) - v^+\left(\frac{n-i-1}{N}\right), \text{ if } 0 \leq i \leq n, \\ v^-\left(\frac{m+i+1}{N}\right) - v^-\left(\frac{m+i}{N}\right), \text{ if } -m \leq i \leq 0, \end{cases} \tag{2.19}$$

with $p_{-m} = v^-\left(\frac{1}{N}\right)$ and $p_n = v^-\left(\frac{1}{N}\right)$.

Therefore, the algorithm to compute the values of the prior weights, given γ^+, δ^+, γ^- and δ^-, is:

1. Sort in ascending order the values of the simulated paths.
2. Find the value closest to the original underlying asset value, so that it becomes the $i = 0$ path.
3. Compute the weights p_i using (2.17) and (2.19).
4. Normalize to 1 by dividing each weight by the sum of all of them.

Figures 2.1 and 2.2 show how the distorsion functions give more weight to the tail in the loss domain at two different maturities for the underlying asset of Banco Santander stock.

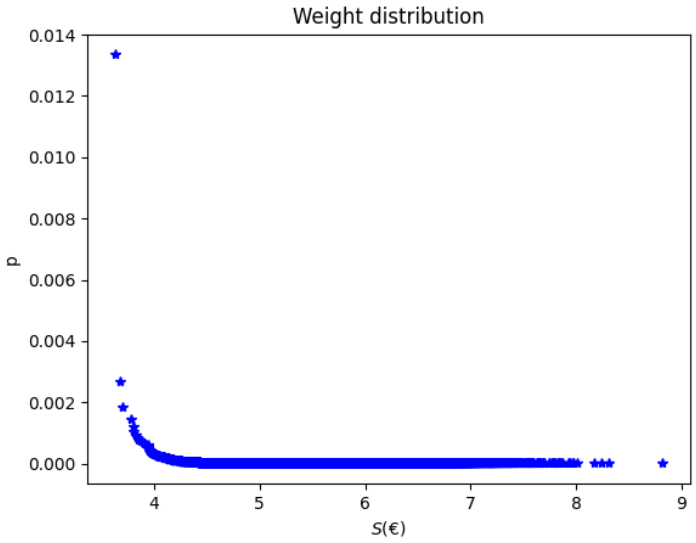


Figure 2.1: Weights distribution for $T = 0.1260$ years with $\gamma^+ = 1.0009$, $\delta^+ = 1.0005$, $\gamma^- = 0.6541$ and $\delta^- = 0.9133$.

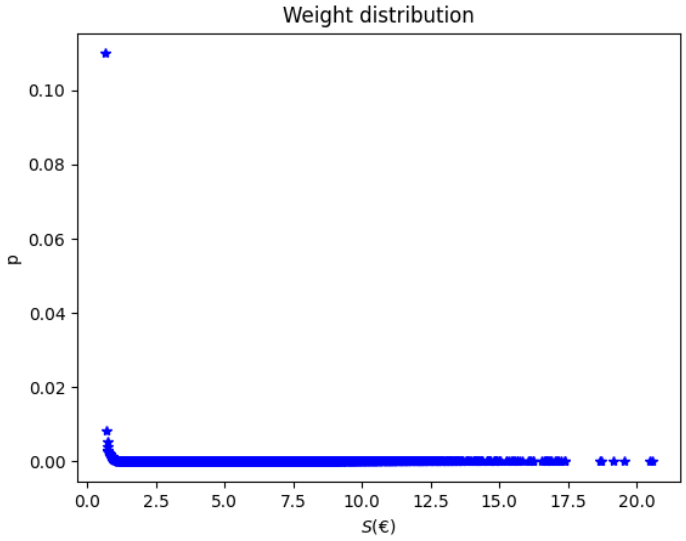


Figure 2.2: Weights distribution for $T = 2.3699$ years with $\gamma^+ = 1.4815$, $\delta^+ = 1.0898$, $\gamma^- = 0.4410$ and $\delta^- = 0.8738$.

2.3.2 Calibration of the parameters γ^+, δ^+, γ^- and δ^-

Let us introduce two error measurements:

- Mean relative price error ($MRPE$):

$$MRPE = \frac{1}{L}\sum_{k=1}^{L}\frac{|\pi_k^{Model} - \pi_k^{Market}|}{\pi_k^{Market}}. \tag{2.20}$$

- Mean average price error ($MAPE$):

$$MAPE = \frac{1}{L}\sum_{k=1}^{L}|\pi_k^{Model} - \pi_k^{Market}|, \tag{2.21}$$

where π_k^{Model} represents the price of option k obtained by our model, and π_k^{Market} represents the real market price of that option. In the calibration procedure, $MAPE$ will not be used. However, it will be employed later in the analysis. Although the definitions of the error measurements are general, we specify that the summations range from $k = 1$ to $k = L$ to emphasize that the K options used in WMC will not be included in the error measurement during calibration.

The first step in GWMC is to apply WMC to K options with any set of parameters. In this way, we obtain the final weights q_i. With them, we calculate the goodness of fit of the model for the L options that were not used in the WMC minimization, as in Equation (2.4) with $k = 1, ..., L$, by computing the value of the $MRPE$ for these L options. We will choose the parameters γ^+, δ^+, γ^- and δ^- of the distortion function, that minimize the $MRPE$. In other words, we solve the following minimization problem:

$$\xi^* = \arg\min_{\xi}\left\{MRPE(\xi)\right\}, \tag{2.22}$$

$$MRPE(\xi) = \frac{1}{L}\sum_{k=1}^{L}\frac{|\pi_k^{Model}(\xi) - \pi_k^{Market}|}{\pi_k^{Market}}, \tag{2.23}$$

where ξ is the set of parameters. This problem must be solved by an optimization solver. We chose the BFGS algorithm for this purpose. It is worth noting that every time the algorithm performs a function evaluation, it calculates the minimization problem (2.8), so the number of options K should

not be too large, as the computational time of Equation (2.15) could become excessive to manage. Also, if all parameters are equal to 1, then we get a uniform distribution of weights, which was used as a starting point for the minimization of the first maturity. For the following maturities, we choose the optimal parameters of the previous one as a starting point.

2.4 Application of the GWMC method

Let us assume that we have an options portfolio in which all benchmark options have the same maturity, then the application of both methods in order to obtain the final weights is straightforward:

1. Simulate N paths of the underlying asset and compute the discounted payoff matrix.

2. If GWMC is being used, obtain the optimal parameters of the distortion function.

3. Make a WMC minimization using the distorted or undistorted prior weights, depending on the method that is being used. By doing this, the values of λ are obtained.

4. Calculate the final weights.

In WMC, it is possible to simulate each path with a value at every maturity we are interested in and treat it as one entity. In other words, it is possible to apply WMC with options with different maturities without having to differentiate among them. The resulting weights will be the probabilities of each whole path being realized.

In GWMC, this is not possible because paths can cross each other between maturites,thus making it impossible to establish a constant order for implementing the distortion. The method proposed in [5] tackles this issue by splitting the set of benchmark options into single-maturity subsets and solving the portfolio problem for each one separately. So, if we have a portofolio of options with M different maturities we will have to solve M single-maturity portofolios.

With both methods, we want to be able to price not only at maturities for which we have options, but also at maturities for which we do not. In WMC, this is pretty simple; we only have to simulate the paths and get their values

at the maturities at which we want to make predictions. Then, we use the weights obtained for the paths. As we have seen, in GWMC we do not have weights for each path but rather a vector of weights for each maturity. In [5], realized values along the simulated paths at times that do not coincide with the maturities present in the set of benchmark options are assigned weights based on linear interpolation with respect to their relative position.

The full scheme proposed by the authors in [5] is as follows. Let J_b and J_s be the two sets of indices such that $\Gamma_b = \{t_j | j \in J_b\}$ is the set of all points in time between 0 and T that coincide with the maturity of a benchmark option, and $\Gamma_s = \{t_j | j \in J_s\}$ is the set of all points in time between 0 and T that do not coincide with a benchmark maturity but for which we would like to know the state price density implied by the benchmark options. Furthermore, given $j \in J_b$, let $S^j = \left\{x^j_{-m}, \ldots, x^j_{-1}, x^j_0, x^j_1, \ldots, x^j_n\right\}$ denote the ordered set of realized points in our sample space at time t_j, such that $x^j_i < x^j_{i'}$ for $i < i'$.

Once the final weights have been computed for each maturity $t_j \in \Gamma_b$, the set of weights for each $t_{j'} \in \Gamma_S$ is calculated as follows. Assume that we have $j, j'' \in J_b$ and $j' \in J_S$ such that $t_j = \sup\{\tau \in \Gamma_b | \tau < t_{j'}\}$ and $t_{j''} = \inf\{\tau \in \Gamma_b | \tau > t_{j'}\}$. Then, we calculate the sorted index arrays I^j, $I^{j'}$, and $I^{j''}$ (i.e., the index arrays for the sorted versions of S^j, $S^{j'}$, and $S^{j''}$). Denote the weight arrays for S^j and $S^{j''}$ by $\mathbf{q}^j$ and $\mathbf{q}^{j''}$, respectively. If $I^{j'}(i) = i'$ we compute:

$$\mathbf{q}^{j'}(i') = \frac{t_{j'} - t_j}{t_{j''} - t_j}\, \mathbf{q}^{\mathbf{j}}(I^j(i)) + \frac{t_{j''} - t_{j'}}{t_{j''} - t_j}\, \mathbf{q}^{\mathbf{j}''}(I^{j''}(i)), \tag{2.24}$$

for $i = 1, \ldots, N$. Finally, if either t_j or $t_{j''}$ falls outside the range of maturities of the benchmark options, then we simply choose whichever benchmark maturity is closest and set the interpolation weight of that maturity to 1.

To summarize:

1. Simulate N paths of the underlying asset and compute the discounted payoff matrix.

2. Obtain the optimal parameters of the distortion function at each maturity.

3. Make a WMC minimization using the distorted prior weights. By doing this, the values of λ are obtained for each maturity.

4. Calculate the final weights at each maturity.

5. For a maturity $t_{j'}$ of interest which does not coincide with the maturities present in the set of benchmark options, use the formula given in (2.24) to interpolate between the two vectors $\mathbf{q}^j$ and $\mathbf{q}^{j''}$ obtained for the two subsets of benchmark options whose maturities t_j and $t_{j''}$ most narrowly sandwich $t_{j'}$. In the event that the maturity is either longer or shorter than anything that is available in the portfolio, attach to it the weight vector computed for the set of benchmark options with the maturity closest to $t_{j'}$.

Chapter 3

Numerical results

In this chapter, we will present the results obtained for European call and put options of three different stocks: Repsol, Banco Santander, and Telefónica. All options are purchased on April 6th, 2013, with 17 maturities ranging from April 17th, 2013, to December 15th, 2017. We will compare the results of both methods, WMC and GWMC.

First, we will explain how the path simulation was carried out and how the options we worked with were selected. Then we will show how the GWMC minimizes both MRPE and MAPE. Finally, we will compare the goodness of fit of both models on maturities at which we have option prices but were not used for the calibrations.

3.1 Path simulation and option selection

We simulated $N = 50000$ paths with the simulations dates equal to the 17 maturities, starting on April 6th, 2013, with the asset following a geometric Brownian motion with volatility dependent on time:

$$dS_t = \mu S_t dt + \sigma_t S_t dW_t, \tag{3.1}$$

where S_t is the asset price, W_t is a Wiener process, σ_t is the volatility and μ is the drift. This stochastic differential equation (SDE) has an analytical solution if σ is constant:

$$S_t = S_0 \exp\left\{\left(\mu - \frac{\sigma^2}{2}\right) t + \sigma W_t\right\}, \tag{3.2}$$

where S_0 is the price of the asset at time $t = 0$. In our model, σ_t changes in each maturity period, being constant from one maturity to the next one. Because of this, we used Equation 3.2 to simulate the new asset price of the path starting from the previous one.

The chosen volatilities for each time step were the implied volatilities by the Black-Scholes model of the option with the closest strike to the forward and the highest price for each maturity. In other words, at maturity t_j, the forward price is F_j, so we choose the option (call or put) with strike price K_j closest to F_j and with the highest price (puts and calls do not have the same value in general). With this option, we calculate the volatility that is needed for the Black-Scholes model to give the market price of the option, and we use it as σ_t in our path generator model.[1]

The selection of the options for which we apply the methods can be a little bit tricky and might require finesse, especially for the ones used in WMC minimizations. This is because the algorithms do not converge if there exists arbitrage in our data. Furthermore, in order to avoid dealing with ill-conditioned matrices, it would be very helpful to not use options with very small prices. Having this in mind, we did not use options with values smaller than 0.5.

We applied the methods for calls and puts separately. In the WMC minimization within GWMC, we chose the five options with strikes closest to the forward price at each maturity and used the rest for the MRPE minimization. If the number of call or put options valued above 0.5 at one maturity was less or equal to 5, we did not use them for the calculations. Also, in the WMC minimization we introduced one more benchmark instrument, the forward, which can be understood as a call option with strike equal to zero.

3.2 Minimization results

In this section, we present the $MRPE$ and $MAPE$ results for the GWMC minimization. That is, we will compare both errors using a uniform distribution of prior weights (WMC) and the distribution given by the optimal parameters of the GWMC minimization.

In Tables A.1, A.2, and A.3, we show the optimal parameters for each

[1] In our dataset, we already had the values of the implied volatilities, which we used to calculate the market prices of the options given by the Black-Scholes formula. The method we just described is the one used to compute these implied volatilities.

maturity and option type for Repsol, Banco Santander and Telefónica, respectively. Empty cells indicate that there were not enough options to apply GWMC, so we also did not apply WMC since we cannot compare results. Cells with * denote that the methods did not converge. This occurred when using options on Telefónica with short maturities, ranging from 11 days to 137 days for put options. Values of options with very short maturities often follow the law of supply and demand rather than model pricing . This makes it difficult to model the price of this kind of options properly and could explain the lack of convergence of the methods for Telefónica options. Furthermore, when all parameters are equal to one, GWMC could not generate a better prior weight distribution than the uniform. This also happens with short maturities.

As we have mentioned before, in GWMC minimizations, we initialize the parameters with the optimal ones from the previous maturity. By analyzing the values of the optimal parameters we can see that, generally, they follow a trend. For example, if we consider Banco Santander call options, then all parameters show a clear trend, either increasing or decreasing with little fluctuations. However, there are some maturities that do not follow any trend. For example, the values of the parameters for $T = 2.3699$ of REPSOL call options show a significant jump. In the remaining maturities, γ^- seems to go back to the previous trend, whereas δ^+ and δ^- take a completely different direction.

In Tables B.1, B.2 and B.3, we show the $MRPE$ and $MAPE$ after the GWMC minimizations. As one would expect, the erros improve in the maturities where the algorithm found an optimal set of parameters different from the one of the uniform distribution. The biggest improvement is 452.73 times better, going from an $MRPE$ of 0.6791% to 0.0015%.

The minimization processes took about 45 minutes for all maturities of each option type and each underlying asset.

3.3 Pricing in other maturities

In this section, we will compare the goodness of fit of both WMC and GWMC on different maturities. For this purpose, we will use the nearest maturities to the one we want to price and calculate the error with respect to the market value.

First, we will use the same five benchmark options from the internal

WMC minimizations of GWMC to obtain the new weights for both methods. Then, we will use all the available options. By doing this, we aim to determine whether it is better to apply WMC leaving some options to calculate a distortion function, i.e., GWMC, or to use a uniform distribution and include every option in the WMC minimization. Also, we will see if using all options in GWMC plus the distortion function is better than using five plus the distortion function.

When applying WMC, we start by using interpolation of weights, and then using the options of both maturities at the same time (as it is intended in the method).

3.3.1 Five benchmark options

GWMC

In Tables C.1, C.2 and C.3 we show the obtained $MRPE$ and $MAPE$ for both calls and puts of REPSOL, Banco Santander and Telefónica, respectively, in all maturities. In Table 3.1 we can see the $MRPE$ and $MAPE$ for all call and put options separately and all of them together for all 3 underlying assets.

	GWMC, 5 BENCHMARK					
	Call		Put		All	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
REP	1.33	2.28	3.61	7.19	2.47	4.71
SAN	1.22	0.96	0.53	0.69	0.84	0.81
TEF	1.14	1.62	1.79	1.93	1.49	1.79

Table 3.1: $MRPE$ and $MAPE$ for all calls, all puts and all options together.

In order to compare the computational cost among all methods, we calculated their time to obtain the $MRPE$ and $MAPE$ for all REPSOL options. In this case it was $t_{GWMC_5} = 38.11$ s.

Interpolating weights in WMC

In Tables C.7, C.8 and C.9 we show $MRPE$ and $MAPE$ values for both call and put options of REPSOL, Banco Santander and Telefónica, respectively, for all maturities. In Table 3.2 we collect both $MRPE$ and $MAPE$ for all

call and put options separately and also all options together for all 3 different stocks. Except for Banco Santander call options case, in which it is slightly better, the fitting is worse than in Table 3.1 for calls and puts; and if we look at the overall errors they are also worse.

	WMC, 5 BENCHMARK, INTERPOLATING					
	Call		Put		All	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
REP	2.15	3.40	4.01	7.89	3.07	5.63
SAN	1.17	0.93	1.40	2.30	1.30	1.69
TEF	1.32	1.88	2.11	2.93	1.74	2.44

Table 3.2: $MRPE$ and $MAPE$ for all calls, all puts and all options together.

The required time to obtain the results for all REPSOL call options was $t_{WMC_{5,interp}} = 18.66$ s.

Not interpolating weights in WMC

In Tables C.10, C.11 and C.12 we show $MRPE$ and $MAPE$ results for call and put options of REPSOL, Banco Santander and Telefónica, respectively, in all maturities. In Table 3.3 we can see $MRPE$ and $MAPE$ values for not differentiating among maturities. These values are greater than those obtained by interpolating weights.

	WMC, 5 BENCHMARK					
	Call		Put		All	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
REP	2.50	3.78	4.29	8.70	3.39	6.23
SAN	1.48	1.20	1.67	2.48	1.58	1.90
TEF	1.41	2.00	2.07	2.98	1.77	2.53

Table 3.3: $MRPE$ and $MAPE$ for all calls, all puts and all options together.

The time it took for the algorithm to achieve the call $MRPE$ and $MAPE$ results with REPSOL stock as the underlying asset was $t_{WMC_5} = 16.14$ s.

3.3.2 All options as benchmark

GWMC

In Tables C.4, C.5 and C.6 we show the results of $MRPE$ and $MAPE$ at each maturity of call and put options of REPSOL, Banco Santander and Telefónica respectively. For Telefónica put options we were not able to price on the maturity $T = 0.6248$ years, as the algorithm did not converge due to the presence of a singular matrix. This problem only appeared for GWMC when using all options as benchmark. Because of this, in Table 3.4, we show the errors for all calls, all puts and all options together for all 3 underlying assets, although not taking into account this maturity for Telefónica puts. Comparing the rest with GWMC using only five benchmark options, the difference between both approaches is minimum, being only slightly better to use all options in some cases, and just the five options on the other ones.

	GWMC, ALL OPTIONS					
	Call		Put		All	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
REP	1.34	2.32	3.61	7.22	2.46	4.75
SAN	1.21	0.95	0.52	0.67	0.83	0.80
TEF	1.15	1.63	2.23	2.23	1.70	2.11

Table 3.4: $MRPE$ and $MAPE$ for all calls, all puts and all options together.

The computational cost of this method is $t_{GWMC_{all}} = 43.18$ s

Interpolating weights in WMC

In Tables C.13, C.14 and C.15 we show the $MRPE$ and $MAPE$ results of calls and puts by maturity for REPSOL, Banco Santander and Telefónica options, respectively. In Table 3.5 we collect the results without differentiating by maturities.

The computational cost to get the $MRPE$ and $MAPE$ for all REPSOL options together was $t_{WMC_{all,interp}} = 33.01$ s.

Not interpolating weights in WMC

In Tables C.16, C.17 and C.18 we can see the values of $MRPE$ and $MAPE$ for all calls, all puts and all options together for REPSOL, Banco Santander

	WMC, ALL OPTIONS , INTERPOLATING					
	Call		Put		All	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
REP	1.37	2.46	3.62	7.17	2.49	4.78
SAN	1.22	0.97	0.37	0.48	0.75	0.7
TEF	1.16	1.65	1.79	2.11	1.50	1.9

Table 3.5: $MRPE$ and $MAPE$ for all calls, all puts and all options together.

and Telefónica, respectively by maturity. In Table 3.6 we show the results for all calls and all puts together, and all options together. These results only improve those in Table 3.5 in the case of Telefónica options.

	WMC, ALL OPTIONS					
	Call		Put		All	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
REP	1.63	2.60	3.75	7.33	2.68	4.95
SAN	1.33	1.08	0.68	0.83	0.97	0.94
TEF	1.11	1.59	1.70	2.13	1.43	1.89

Table 3.6: $MRPE$ and $MAPE$ for all calls, all puts and all options together.

The computational cost of calculating the errors for all REPSOL options was $t_{WMC_{all}} = 28.92$ s.

3.3.3 Results analysis

Let us start by comparing the results obtained by GWMC with five benchmark options and all options as benchmark. If we take a look at the results in Tables 3.1 and 3.4 for REPSOL and Banco Santander, for both option types, separately and together, the errors barely change from using five to all options as benchmark. For REPSOL options, $MRPE$ decreases by 0.01% and $MAPE$ grows by 0.04% (calls and all options) and 0.03% (puts). For Banco Santander, all errors decrease by either 0.01% or 0.02%. The problem comes with Telefónica puts, because, as we have mentioned before, having a high number of benchmark options can lead to a bad behavior of the method motivated by the appearance of some kind of arbitrage in our data. Indeed, this happened with Telefónica put options for the maturity at $t = 0.6248$ years. For call options, $MRPE$ and $MAPE$ both grow by 0.01% , which is a very

small amount. If we apply GWMC with five benchmark options without this maturity for Telefónica put options, the results are:

$$MRPE_{puts} = 2.25\% \quad MAPE_{puts} = 2.29\%$$
$$MRPE_{all} = 1.70\% \quad MAPE_{all} = 1.95\%,$$

which, again, are roughly the same as using all options as benchmark. Taking all this into account, and also the fact that the computation cost is lower, we deduce that it is better to apply GWMC in order to make predictions with few options as benchmark.

Comparing between applying WMC interpolating weights or how it is supposed to be applied (not interpolating weights), we can see that interpolating gives better results except for Telefónica options when all options are used as benchmark. We cannot conclude that interpolating weights will be better in general, but further investigations with more data may be needed in order to clarify these results.

In view that WMC interpolating weights gave us better results, let us compare them with GWMC using five benchmark options. If we also use 5 options as benchmark, GWMC is better in every case. However, if we apply WMC with all options as benchmark options, i.e., if we give the WMC model the same information as to the GWMC model (even if we use five benchmark options it has the information acquired from the rest via the distortion function), the new errors come really close to those obtained by GWMC. A possible explanation for this is that the information the WMC model gets from all options , when used for pricing in other maturities, translates in a similar way as fitting using the distortion function. In other words, WMC gets from all options roughly the same information as GWMC from 5 plus the distortion function.

All in all, from this study we may conclude that, between WMC and GWMC, the best model to make predictions would be GWMC using five options as benchmark. More precisely, eventhough the values of $MRPE$ and $MAPE$ of WMC interpolating weights are almost identical to the ones we got from GWMC without the extra computational time that is needed to obtain the distortion functions, using a big amount of options might lead to not converging algorithms.

Chapter 4

Conclusions and further lines of research

We have performed a study comparing two methodologies for option pricing, Weighted Monte Carlo and Generalized Weighted Monte Carlo, using data from REPSOL, Banco Santander and Telefónica European call and put options with maturities ranging from 11 days to 4 and a half years. The results showed that the distortion function from GWMC gathers approximately the same information as all the options combined. In view of these results and to take advantage of this, GWMC is preferable to WMC with a large number of options in order to prevent the model from failing. Furthermore, if we had very few options and another way to get to the distorsion function, GWMC would be able to use this extra information and might be able to achieve better results. Also, with our data, when using WMC it is better to interpolate weights in order to price options in other maturities rather than to calculate the weight of the whole path and using it to make the prediction.

Different extensions of this work could be to keep applying both methods to generalize these results and to explore if interpolating weights in WMC is, in fact, better or if it was just a coincidence. Other further research could consist on the estimation of the distortion function without using option values and analysing if that information also helps GWMC to make better predictions.

Appendix A
Minimization parameters

REPSOL								
T(years)	Call				Put			
	γ^+	δ^+	γ^-	δ^-	γ^+	δ^+	γ^-	δ^-
0.0301	1.0000	1.0000	1.0000	1.0000	–	–	–	–
0.1260	1.0006	1.0001	0.7895	0.9423	1.1993	1.0408	0.9965	0.9973
0.2027	1.0007	1.0001	0.7759	0.9389	1.2630	1.0587	0.9986	0.9990
0.3753	1.3187	1.076	0.7385	0.9411	1.3161	1.0689	0.7617	0.9073
0.6248	1.4031	1.1008	0.6767	0.9255	1.4245	1.1001	0.6650	0.8800
0.8741	1.1008	1.0185	0.6530	0.9248	1.1966	1.0372	0.6508	0.8812
1.1233	1.0913	1.0182	0.6568	0.9271	1.3092	1.0587	0.6928	0.8889
1.3726	1.085	1.0180	0.6544	0.9273	1.1766	1.0488	0.6082	0.8874
1.6220	1.1444	1.0258	0.6153	0.9151	1.1764	1.0492	0.5687	0.8765
1.8713	1.1589	1.0284	0.6124	0.9145	1.2192	1.0603	0.5641	0.8755
2.1205	1.1958	1.0311	0.5955	0.9074	1.2857	1.0693	0.5670	0.8749
2.3699	1.5351	1.6581	1.2362	1.7688	1.3947	1.7119	0.5438	1.1574
2.6193	0.7834	1.5529	0.4185	1.6037	1.3947	1.7119	0.5438	1.1574
3.1178	0.8105	1.5485	0.4183	1.6037	1.2879	1.7278	0.5855	1.1682
3.6166	0.5997	1.6354	0.6674	1.6057	1.2879	1.7278	0.5855	1.1682
4.1151	0.5997	1.6354	0.6674	1.6057	1.2879	1.7278	0.5855	1.1682
4.6138	0.5518	1.6621	0.7034	1.6127	1.2879	1.7278	0.5855	1.1682

Table A.1: Values of the parameters γ^+, δ^+, γ^-, and δ^- for REPSOL call and put options for each maturity. If there are no values for one maturity and option type, it is because there were not enough options to apply the method with.

BANCO SANTANDER								
T(years)	Call				Put			
	γ^+	δ^+	γ^-	δ^-	γ^+	δ^+	γ^-	δ^-
0.0301	1.0000	1.0000	1.0000	1.0000	–	–	–	–
0.1260	1.0009	1.0005	0.6541	0.9133	1.0000	1.0000	1.0000	1.0000
0.2027	1.0009	1.0005	0.6427	0.9108	–	–	–	–
0.3753	1.0010	1.0005	0.6231	0.9062	1.0000	1.0000	1.0000	1.0000
0.6248	1.0011	1.0005	0.5504	0.8881	1.0000	1.0000	1.0000	1.0000
0.8741	1.0012	1.0005	0.5175	0.8800	1.6623	1.1175	0.9609	0.9506
1.1233	1.0012	1.0005	0.5122	0.8786	1.4227	1.0637	0.9975	0.9945
1.3726	1.3348	1.0452	0.5231	0.8878	1.3909	0.7167	0.4347	1.4524
1.6220	1.3647	1.0533	0.5003	0.8828	1.4337	0.7345	0.3841	1.4516
1.8713	1.3526	1.0497	0.5065	0.8838	1.4409	0.7380	0.3786	1.4518
2.1205	1.4575	1.0825	0.4533	0.8757	1.4962	0.7601	0.3205	1.4531
2.3699	1.4815	1.0898	0.4411	0.8738	1.5224	0.776	0.3194	1.4527
2.6193	–	–	–	–	1.8574	1.695	1.5580	1.6051

Table A.2: Values of the parameters γ^+, δ^+, γ^-, and δ^- for Banco Santander call and put options for each maturity.

TELEFÓNICA								
T(years)	Call				Put			
	γ^+	δ^+	γ^-	δ^-	γ^+	δ^+	γ^-	δ^-
0.0301	*	*	*	*	*	*	*	*
0.1260	1.0000	1.0000	1.0000	1.0000	*	*	*	*
0.2027	1.0020	1.0007	0.6771	0.9174	*	*	*	*
0.3753	1.0020	1.0007	0.6670	0.915	*	*	*	*
0.6248	1.0021	1.0007	0.6670	0.915	1.4614	1.0811	0.9994	0.9994
0.8741	1.0018	1.0008	0.5764	0.8924	1.4716	1.0499	0.5349	0.8510
1.1233	0.9805	0.9977	0.5653	0.8896	1.4119	1.0296	0.5539	0.8572
1.3726	0.7744	0.9727	0.5544	0.8874	1.3878	1.022	0.5348	0.8522
1.6220	1.1124	0.9765	0.9215	0.9391	1.3469	1.0185	0.9134	0.9662
1.8713	1.1328	0.9794	0.9224	0.9385	1.2782	1.0639	0.8600	1.0784
2.1205	1.1333	0.9797	0.8990	0.9321	1.2802	1.0664	0.8097	1.064
2.3699	1.1356	0.9820	0.8584	0.9257	1.2854	1.0679	0.8011	1.0615
2.6193	1.2668	0.7851	1.1277	1.1187	1.2665	1.0681	0.8086	1.0663

Table A.3: Values of the parameters γ^+, δ^+, γ^-, and δ^- for Telefónica call and put options for each maturity. If for one maturity and option type tear are * in each cell, then the methods could not be applied.

Appendix B

Same maturity pricing

REPSOL								
T(years)	Call				Put			
	$MRPE(\%)$		$MAPE(\%)$		$MRPE(\%)$		$MAPE(\%)$	
	WMC	GWMC	WMC	GWMC	WMC	GWMC	WMC	GWMC
0.0301	0.0029	0.0029	0.0092	0.0092	–	–	–	–
0.1260	0.1573	0.0017	0.5024	0.0068	0.1337	0.0027	0.4007	0.0097
0.2027	0.3388	0.0026	0.9486	0.0068	0.4717	0.0093	1.2644	0.0228
0.3753	0.7657	0.0172	2.5330	0.0622	0.6598	0.0238	1.8130	0.0716
0.6248	1.5433	0.0366	3.5198	0.0995	1.5436	0.0490	3.0807	0.2240
0.8741	1.0587	0.3079	2.1922	0.2515	2.6897	0.1729	2.9348	0.6577
1.1233	1.8571	0.3259	2.2959	0.2917	1.8502	0.2408	3.7974	0.8936
1.3726	1.6992	0.2812	2.3774	0.3194	2.5600	0.2136	4.3395	0.9009
1.6220	2.5769	0.6700	3.6370	0.6501	3.1843	0.2331	5.0036	1.0141
1.8713	3.5789	0.6586	3.6802	0.6375	1.4496	0.2592	5.5933	1.3021
2.1205	5.4828	0.5340	5.3124	0.4885	1.6159	0.1961	7.6538	1.0862
2.3699	7.2540	0.3596	6.8337	0.6815	1.8032	0.1921	8.7288	1.0892
2.6193	1.0912	0.2596	4.3136	0.4094	1.0208	0.0667	4.4794	0.2702
3.1178	0.5933	0.3504	0.8203	0.5054	0.1146	0.0998	0.6102	0.5144
3.6166	0.7073	0.3235	1.1194	0.5020	0.1221	0.1090	0.6565	0.5699
4.1151	0.9403	0.3173	1.3513	0.4760	0.1161	0.1033	0.7104	0.6287
4.6138	0.9606	0.2481	1.5528	0.4243	0.1111	0.1010	0.6830	0.6180

Table B.1: Errors for REPSOL options on the same maturity.

BANCO SANTANDER								
T(years)	Call				Put			
	$MRPE(\%)$		$MAPE(\%)$		$MRPE(\%)$		$MAPE(\%)$	
	WMC	GWMC	WMC	GWMC	WMC	GWMC	WMC	GWMC
0.0301	0.0016	0.0016	0.0017	0.0017	–	–	–	–
0.1260	0.0598	0.0063	0.0774	0.0091	0.0041	0.0041	0.0075	0.0075
0.2027	0.1327	0.0008	0.1350	0.0008	–	–	–	–
0.3753	0.2689	0.0036	0.3245	0.0042	0.0767	0.0767	0.1539	0.1539
0.6248	0.5332	0.0048	0.6121	0.0050	0.2255	0.2255	0.5776	0.5776
0.8741	0.5989	0.0106	0.6745	0.0106	1.0550	0.0332	1.5623	0.0545
1.1233	0.6159	0.0085	0.6672	0.0084	1.6091	0.2280	2.3580	0.3135
1.3726	0.6462	0.0111	0.6075	0.0115	2.0057	0.2591	3.1263	0.4532
1.6220	0.8264	0.0098	0.4874	0.0091	2.2388	0.2312	3.6789	0.4136
1.8713	1.0737	0.0260	0.6071	0.0185	2.2987	0.1643	3.9929	0.3134
2.1205	1.011	0.0159	0.6464	0.0112	2.7176	0.2572	5.2693	0.5446
2.3699	0.9850	0.0203	0.6144	0.0136	2.4304	0.1772	4.779	0.3567
2.6193	–	–	–	–	0.3548	0.0727	0.8505	0.1773

Table B.2: Errors of Banco Santander options on the same maturity.

TELEFÓNICA								
T(years)	Call				Put			
	$MRPE(\%)$		$MAPE(\%)$		$MRPE(\%)$		$MAPE(\%)$	
	WMC	GWMC	WMC	GWMC	WMC	GWMC	WMC	GWMC
0.1260	0.0184	0.0184	0.0447	0.0447	*	*	*	*
0.2027	0.0688	0.0020	0.1538	0.0042	*	*	*	*
0.3753	0.2812	0.0036	0.7042	0.0081	*	*	*	*
0.6248	0.4992	0.0078	1.1591	0.0164	0.2488	0.1011	1.1360	0.6009
0.8741	0.9282	0.0179	2.2158	0.0418	0.7780	0.1014	2.5297	0.4311
1.1233	0.6774	0.0055	1.5021	0.0117	0.8597	0.1983	2.5012	0.8770
1.3726	0.6791	0.0015	1.4992	0.0035	1.8861	0.1191	1.9249	0.2993
1.6220	0.3735	0.0305	0.4361	0.0304	0.7823	0.2889	3.9199	1.6852
1.8713	0.6288	0.0861	0.5042	0.0736	0.7228	0.1290	1.7982	0.3467
2.1205	1.2140	0.1271	0.9254	0.1381	0.8924	0.1493	2.6383	0.4349
2.3699	1.3657	0.0911	1.0555	0.0746	0.9643	0.1363	2.9710	0.433
2.6193	1.3588	0.0359	1.0482	0.0313	0.5721	0.0371	1.6008	0.0971

Table B.3: Errors of Telefónica options on the same maturity.

Appendix C
Pricing in other maturities

C.1 GWMC

C.1.1 Five benchmark options

REPSOL				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.0301	0.71	0.80	–	–
0.1260	0.35	0.59	0.87	1.27
0.2027	0.38	0.64	0.88	0.65
0.3753	0.28	0.49	0.64	0.77
0.6248	0.52	0.67	0.25	0.64
0.8741	0.29	0.36	7.36	10.30
1.1233	1.43	2.30	14.89	22.73
1.3726	0.23	0.32	6.35	11.48
1.6220	1.85	2.80	2.24	4.27
1.8713	1.02	1.75	3.19	8.39
2.1205	1.32	2.28	3.11	8.73
2.3699	1.38	1.20	2.47	7.73
2.6193	3.54	6.94	2.38	7.66
3.1178	3.63	6.56	1.13	4.84
3.6166	3.81	7.59	0.91	4.09
4.1151	3.90	7.38	1.00	5.02
4.6138	1.56	3.19	0.37	2.02

Table C.1: Errors of GWMC applied to REPSOL call and put options.

BANCO SANTANDER				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.0301	0.46	0.32	–	–
0.1260	0.22	0.17	–	–
0.2027	2.32	1.68	–	–
0.3753	1.77	1.57	0.80	0.74
0.6248	0.43	0.31	0.68	1.04
0.8741	0.99	0.85	0.54	0.40
1.1233	2.26	1.83	0.12	0.17
1.3726	0.14	0.13	0.49	0.54
1.6220	0.35	0.28	0.21	0.32
1.8713	1.37	1.06	0.44	0.54
2.1205	0.95	0.74	0.34	0.52
2.3699	3.74	2.84	0.52	0.60
2.6193	–	–	1.97	3.10

Table C.2: Errors of GWMC applied to Banco Santander call and put options.

TELEFÓNICA				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.1260	1.34	1.35	*	*
0.2027	0.45	0.43	*	*
0.3753	0.59	0.52	*	*
0.6248	0.46	0.5	0.42	1.08
0.8741	0.93	0.90	1.19	1.68
1.1233	1.47	1.86	1.77	2.16
1.3726	2.86	5.27	7.02	5.36
1.6220	2.92	5.13	2.79	2.95
1.8713	0.13	0.22	0.26	0.38
2.1205	0.48	0.49	0.99	1.32
2.3699	0.66	0.72	0.15	0.36
2.6193	1.17	1.34	0.85	1.27

Table C.3: Errors of GWMC applied to Telefónica call and put options.

C.1.2 Using all options as benchmark

REPSOL				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.0301	0.71	0.79	–	–
0.1260	0.32	0.44	0.85	1.16
0.2027	0.39	0.66	0.87	0.63
0.3753	0.27	0.46	0.64	0.74
0.6248	0.60	0.87	0.17	0.30
0.8741	0.31	0.36	7.34	10.37
1.1233	1.46	2.20	14.84	22.41
1.3726	0.18	0.24	6.37	11.72
1.6220	1.74	2.78	2.24	4.31
1.8713	0.83	1.52	3.15	8.05
2.1205	1.42	2.22	3.19	9.35
2.3699	0.97	0.94	2.52	8.07
2.6193	3.85	8.38	2.38	7.71
3.1178	3.77	6.71	1.11	4.69
3.6166	3.95	7.88	0.94	4.24
4.1151	4.09	7.62	0.99	4.99
4.6138	1.58	3.18	0.40	2.21

Table C.4: Errors of GWMC applied to REPSOL call and put options.

BANCO SANTANDER				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.0301	0.45	0.31	–	–
0.1260	0.21	0.16	–	–
0.2027	2.33	1.69	–	–
0.3753	1.73	1.51	0.93	0.95
0.6248	0.42	0.30	0.82	1.44
0.8741	1.01	0.86	0.53	0.36
1.1233	2.26	1.83	0.10	0.13
1.3726	0.13	0.12	0.47	0.43
1.6220	0.35	0.28	0.08	0.07
1.8713	1.37	1.06	0.43	0.50
2.1205	0.95	0.74	0.29	0.38
2.3699	3.74	2.84	0.49	0.51
2.6193	–	–	2.01	3.18

Table C.5: Errors of GWMC for Banco Santander call and put options.

TELEFÓNICA				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.1260	1.31	1.27	*	*
0.2027	0.46	0.44	*	*
0.3753	0.62	0.59	*	*
0.6248	0.45	0.47	*	*
0.8741	0.93	0.91	3.37	3.44
1.1233	1.47	1.86	1.74	2.05
1.3726	2.87	5.28	6.98	5.3
1.6220	2.9	5.12	2.81	3.11
1.8713	0.11	0.19	0.21	0.23
2.1205	0.47	0.48	0.98	1.23
2.3699	0.81	0.98	0.14	0.36
2.6193	1.14	1.32	0.76	1.05

Table C.6: Errors of GWMC applied to Telefónica call and put options.

C.2 WMC

C.2.1 Five benchmark options

Interpolating weights

REPSOL				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.0301	0.68	0.77	–	–
0.1260	0.40	0.68	0.64	0.70
0.2027	0.38	0.72	0.97	1.10
0.3753	0.55	1.57	0.73	1.64
0.6248	1.29	2.67	0.96	1.66
0.8741	1.42	2.04	8.45	10.09
1.1233	1.49	2.79	14.54	23.85
1.3726	1.23	1.72	7.56	11.75
1.6220	3.32	4.39	3.92	5.60
1.8713	2.32	2.93	3.49	10.82
2.1205	2.93	3.48	2.77	7.51
2.3699	3.50	3.60	3.32	12.05
2.6193	4.54	9.42	2.01	7.27
3.1178	3.94	6.96	0.84	3.37
3.6166	3.74	7.48	0.89	4.00
4.1151	4.50	8.27	0.90	4.43
4.6138	1.35	2.85	0.56	3.05

Table C.7: Errors of WMC applied to Repsol call and put options.

BANCO SANTANDER				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.0301	0.47	0.34	–	–
0.1260	0.28	0.24	–	–
0.2027	2.59	1.92	–	–
0.3753	1.24	0.99	0.80	0.74
0.6248	0.71	0.57	0.91	1.44
0.8741	1.36	1.25	1.08	1.11
1.1233	2.06	1.59	0.99	1.38
1.3726	0.50	0.47	1.43	2.21
1.6220	0.49	0.37	1.56	2.51
1.8713	1.12	0.88	1.73	2.96
2.1205	1.17	0.86	1.78	3.45
2.3699	2.37	1.93	1.45	2.48
2.6193	–	–	1.96	3.73

Table C.8: Errors of WMC applied to Banco Santander call and put options.

TELEFÓNICA				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.1260	1.27	1.23	*	*
0.2027	0.6	0.72	*	*
0.3753	0.87	1.00	*	*
0.6248	0.63	0.86	0.59	1.69
0.8741	1.18	1.56	1.10	2.21
1.1233	1.83	2.63	1.99	2.96
1.3726	3.08	5.74	7.67	6.21
1.6220	2.71	4.73	3.10	4.34
1.8713	0.57	0.59	0.81	1.54
2.1205	0.63	0.63	1.39	2.83
2.3699	1.21	1.08	0.57	1.79
2.6193	0.90	1.02	0.97	1.49

Table C.9: Errors of WMC applied to Telefónica call and put options.

Not interpolating weights

REPSOL				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.0301	0.25	0.46	–	–
0.1260	0.45	0.69	0.72	0.64
0.2027	0.77	1.39	1.13	1.87
0.3753	1.53	2.71	1.60	2.95
0.6248	2.39	3.73	2.00	3.26
0.8741	2.78	3.56	8.96	10.03
1.1233	1.35	2.34	14.93	24.63
1.3726	1.75	2.41	7.76	11.73
1.6220	3.68	4.73	3.64	5.07
1.8713	3.24	4.02	3.7	11.61
2.1205	2.86	3.69	2.45	6.25
2.3699	3.51	4.14	3.18	11.63
2.6193	4.42	8.48	1.84	6.80
3.1178	0.96	2.03	0.78	3.68
3.6166	6.42	12.10	1.73	8.44
4.1151	0.98	1.95	0.70	3.86
4.6138	4.75	8.53	2.81	15.84

Table C.10: Errors of WMC applied to Telefónica call and put options.

BANCO SANTANDER				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.0301	0.23	0.21	–	–
0.1260	0.45	0.35	–	–
0.2027	3.66	2.70	–	–
0.3753	0.60	0.62	0.74	0.84
0.6248	1.23	1.17	1.44	2.09
0.8741	1.71	1.65	1.91	2.03
1.1233	2.51	1.85	1.78	2.13
1.3726	0.77	0.64	1.52	2.10
1.6220	0.89	0.72	1.74	2.40
1.8713	1.02	0.75	2.15	3.25
2.1205	1.82	1.32	1.93	3.40
2.3699	3.19	2.68	1.09	1.87
2.6193	–	–	2.26	4.41

Table C.11: Errors of WMC applied to Banco Santander call and put options.

TELEFÓNICA				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.1260	0.75	0.60	*	*
0.2027	1.46	1.78	*	*
0.3753	1.21	1.52	*	*
0.6248	0.89	1.45	0.93	2.62
0.8741	1.86	2.12	1.52	2.90
1.1233	1.46	2.58	2.00	2.68
1.3726	3.43	6.13	7.67	6.22
1.6220	1.98	3.56	2.22	3.39
1.8713	0.68	0.77	0.82	1.59
2.1205	0.65	0.71	1.35	2.76
2.3699	1.19	1.01	0.69	1.86
2.6193	1.37	1.48	1.35	2.46

Table C.12: Errors of WMC applied to Telefónica options.

C.2.2 Using all options as benchmark

Interpolating weights

REPSOL				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.0301	0.71	0.79	–	–
0.1260	0.45	1.02	0.92	1.49
0.2027	0.38	0.63	0.82	0.60
0.3753	0.26	0.50	0.55	0.62
0.6248	0.77	1.21	0.22	0.46
0.8741	0.34	0.38	7.50	10.51
1.1233	1.45	2.18	15.00	22.64
1.3726	0.18	0.25	6.56	12.04
1.6220	1.82	3.19	2.55	4.52
1.8713	0.84	1.52	2.94	7.53
2.1205	1.42	2.22	3.16	9.27
2.3699	0.93	0.90	2.35	7.04
2.6193	3.9	8.51	1.71	6.01
3.1178	3.78	6.72	1.32	5.71
3.6166	3.91	7.78	0.90	4.11
4.1151	4.13	7.69	0.98	4.97
4.6138	1.57	3.16	0.59	3.25

Table C.13: Errors of WMC applied to Telefónica options.

BANCO SANTANDER				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.0301	0.45	0.31	–	–
0.1260	0.35	0.36	–	–
0.2027	2.35	1.69	–	–
0.3753	1.85	1.67	0.93	0.95
0.6248	0.66	0.50	0.69	1.21
0.8741	0.96	0.82	0.52	0.36
1.1233	2.32	1.88	0.13	0.12
1.3726	0.16	0.13	0.19	0.17
1.6220	0.25	0.21	0.11	0.10
1.8713	1.28	0.99	0.09	0.13
2.1205	0.92	0.71	0.07	0.12
2.3699	3.51	2.68	0.38	0.41
2.6193	–	–	1.55	2.42

Table C.14: Errors of WMC applied to Banco Santander call and put options.

TELEFÓNICA				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.1260	1.38	1.52	*	*
0.2027	0.45	0.42	*	*
0.3753	0.79	0.77	*	*
0.6248	0.47	0.51	0.29	0.35
0.8741	0.96	0.98	0.95	1.21
1.1233	1.46	1.83	1.70	2.44
1.3726	2.87	5.28	6.99	5.29
1.6220	2.90	5.11	3.16	5.05
1.8713	0.11	0.19	0.26	0.26
2.1205	0.45	0.46	0.97	1.20
2.3699	0.74	0.86	0.08	0.12
2.6193	1.09	1.26	0.71	0.97

Table C.15: Errors of WMC applied to Telefónica call and put options.

Not interpolating weights

REPSOL				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.0301	0.25	0.46	–	–
0.1260	0.51	1.24	0.78	0.82
0.2027	0.43	0.64	0.86	1.32
0.3753	0.95	1.20	0.99	1.74
0.6248	1.54	1.59	0.89	1.68
0.8741	1.63	1.57	8.11	10.52
1.1233	0.54	1.07	15.55	23.61
1.3726	0.68	0.86	6.63	11.83
1.6220	2.27	3.05	2.28	3.41
1.8713	0.98	1.30	3.31	8.76
2.1205	0.89	2.02	2.73	7.57
2.3699	2.27	2.48	2.34	7.28
2.6193	2.98	6.03	1.28	4.69
3.1178	1.22	2.59	0.49	2.03
3.6166	5.74	10.95	1.57	7.48
4.1151	1.09	2.37	0.27	1.29
4.6138	6.15	11.54	1.89	10.65

Table C.16: Errors of WMC applied to REPSOL call and put options

BANCO SANTANDER				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.0301	0.20	0.17	–	–
0.1260	0.94	0.93	–	–
0.2027	3.51	2.56	–	–
0.3753	0.28	0.28	0.61	0.65
0.6248	0.52	0.43	1.05	1.58
0.8741	1.36	1.22	1.19	1.16
1.1233	2.44	1.93	1.02	0.96
1.3726	0.31	0.24	0.40	0.42
1.6220	0.92	0.77	0.52	0.47
1.8713	1.04	0.86	0.59	0.77
2.1205	0.75	0.56	0.41	0.56
2.3699	4.14	3.29	0.54	0.78
2.6193	–	–	0.86	1.28

Table C.17: Errors of WMC applied to Banco Santander call and put options.

TELEFÓNICA				
T(years)	Call		Put	
	$MRPE(\%)$	$MAPE(\%)$	$MRPE(\%)$	$MAPE(\%)$
0.0301	*	*	*	*
0.1260	0.26	0.30	*	*
0.2027	1.36	1.59	*	*
0.3753	0.88	0.81	*	*
0.6248	0.39	0.47	0.60	1.44
0.8741	1.39	1.14	1.42	2.03
1.1233	1.18	1.87	1.47	1.73
1.3726	3.38	5.98	6.92	5.28
1.6220	1.86	3.40	2.29	4.17
1.8713	0.26	0.39	0.27	0.37
2.1205	0.47	0.56	1.00	1.28
2.3699	0.74	0.76	0.19	0.31
2.6193	0.87	1.15	0.72	1.05

Table C.18: Errors of WMC applied to Telefónica call and put options.

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